Printed Pages – 6 Roll No. : .....

# 300112(11)

## B.E. (First & Second Semester) Examination, 2020

**(Old Scheme)**

**(Branch : AEI, Bio Tech, Chem., Civil, CSE, Elect., EEE, EI, ET & T, IT, Mech., Mining, Metallurgy, Mechatronics, Production, Automobile, Agriculture, Plastic & Lateral Diploma)**

## APPLIED CHEMISTRY

***Time Allowed : Three hours***

***Maximum Marks : 80***

***Minimum Pass Marks : 28***

***Note :*** *Attempt all questions. Part (a) of each question is compulsory which is of 2 marks. Attempt any two parts from the rest three (b), (c) and (d), each is of 7 marks.*

### Unit - I

1. (a) Why the complex of $Ca^{++}$ or $Mg^{++}$ with EDTA is more stable than the complex of $Ca^{++}$ or $Mg^{++}$ with EBT? 2

(b) A water sample gave the following constituents on analysis : $Ca(HCO_3)_2$ = 8.1 ppm, $CaCl_2$ = 33·3 ppm, $HCO_3$= 91·5 ppm, $Mg(HCO_3)_2$ = 14·6 ppm, $MgCl_2$ =38 ppm, $MgSO_4$ = 12 ppm. The coagulant $Al_2(SO_4)_3$ was added at the rate of 71·1 mg/l of water. Calculate the cost of the chemicals required for softening one million liters of water if the purity of lime is 90% and that of soda is 98%. The costs per kg of lime and soda are ₹ 14 and ₹ 48 respectively. 7

(c) (i) The hardness of 10,000 litres of a sample of water (containing 341·9 ppm hardness) was completely removed by passing it through a zeolite softener. The zeolite softner was regenerated by passing sodium chloride solution containing 20 gms/l of sodium chloride. How many litres of sodium chloride will be required to regenerate zeolite softener? 3½

(ii) What is break point chlorination? What are its advantages? 3½

(d) (i) Write the difference between boiler scales and sludges. 3½

(ii) What is meant by exhaustion of cation and anion exchangers? How can they be regenerated? 3½

**Unit-II**

2. (a) What are the corrections to be made in the calorific value of a fuel, determined by Bomb Calorimeter? 2

(b) 1000 $m^3$ of a producer gas having the volumetric composition CO = 22%, $CO_2$ = 4%, $CH_4$ = 2%, $H_2$ = 14%, $N_2$ = 56%, $O_2$ = 2% has to be burnt for heating a furnace.

(i) Calculate the minimum quantity of air required for the complete combustion.

(ii) What is the volumetric composition of dry flue gas? 7

(c) Write short notes on any two of the following : 3½+3½

(i) Gross calorific value and net calorific value

(ii) Low temperature carbonization and high temperature carbonization

(iii) Octane number and cetane number

(d) (i) 1.0 gm of a sample of coal was used for nitrogen estimation by kjeldahl method. The evolved ammonia was collected in 25 ml

(N/10) sulphuric acid. To neutralize excess acid, 15 ml of 0.1 N sodium hydroxide was required. Determine the percentage of nitrogen in the given sample of coal. 2

(ii) Describe Otto-Hoffmann's process for preparing Metallurgical coke and recovery of byproducts. 5

## Unit-III

3. (a) Which of the following is not on environment modification technique to prevent corrosion : 2

(i) Dehumidification

(ii) Deaeration

(iii) Neutralization of acidic environment

(iv) Using anodic inhibitors

(b) Explain the mechanism of hydrogen evolution and oxygen absorption in wet corrosion. Give figures. 7

(c) (i) Write the discharging and charging reactions of lead acid storage cell? 3½

(ii) Discuss sacrificial anodic protection. What will be the condition for a metal to act as a sacrificial anode to iron? 3½

(d) Write short notes on any **two** of the following : 3½+3½

(i) Oxidation corrosion

(ii) Water line corrosion

(iii) Fuel cell

## Unit - IV

4. (a) Why is the Teflon highly chemical resistant? 2

(b) (i) What is polyamide? Discuss various types of Nylons. 3½

(ii) Discuss the vulcanization of rubber and its advantages. 3½

(c) Write short notes on the following : 3½+3½

(i) Extreme pressure lubricant

(ii) Solid lubricants

(d) Name the basic constitutes of portland cement. What is their average percentage in the cement? Discuss setting and hardening of the cement. 7

## Unit - V

5. (a) Write the the preparation of Melamine. 2

(b) Discuss the preparaction, properties and uses of any **two** commercially important chemicals : 7

(i) Glycerine

(ii) Ammonium sulphate

(iii) Ethyl Ether

(iv) Bromine

(c) Write short notes on any **two** of the following : 7

(i) Dynamite

(ii) Classification of explosive

(iii) RDX

(d) (i) What are the basic requirements of Rocket Propellants? 3½

(ii) How are rocket propellants classified? 3½